

Kinetics of Hydride Transfer from Catalytic Metal-Free Hydride Donors to CO₂

Ravindra B. Weerasooriya, Jonathan L. Gesiorski, Abdulaziz Alherz, Stefan Ilic, George N. Hargenrader, Charles B. Musgrave, and Ksenija D. Glusac*

Cite This: *J. Phys. Chem. Lett.* 2021, 12, 2306–2311

Read Online

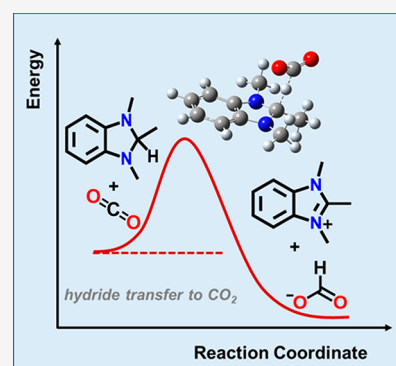
ACCESS |

Metrics & More

Article Recommendations

Supporting Information

ABSTRACT: Selective reduction of CO₂ to formate represents an ongoing challenge in photoelectrocatalysis. To provide mechanistic insights, we investigate the kinetics of hydride transfer (HT) from a series of metal-free hydride donors to CO₂. The observed dependence of experimental and calculated HT barriers on the thermodynamic driving force was modeled by using the Marcus hydride transfer formalism to obtain the insights into the effect of reorganization energies on the reaction kinetics. Our results indicate that even if the most ideal hydride donor were discovered, the HT to CO₂ would exhibit sluggish kinetics (<100 turnovers per second at −0.1 eV driving force), indicating that the conventional HT may not be an appropriate mechanism for solar conversion of CO₂ to formate. We propose that the conventional HT mechanism should not be considered for CO₂ reduction catalysis and argue that the orthogonal HT mechanism, previously proposed to address thermodynamic limitations of this reaction, may also lead to lower kinetic barriers for CO₂ reduction to formate.



The conversion of CO₂ to value-added products is a promising approach to mitigate the effect of increasing levels of CO₂ in the atmosphere. The two-electron proton-coupled reduction of CO₂ to formic acid represents an attractive transformation because the product can serve as a fuel with high volumetric energy density, easy handling, and low toxicity or as a reagent in the pharmaceutical industry and other synthesis applications.^{1,2} However, the selective reduction of CO₂ to formate remains a challenge, and most transition metals catalyze the competing reactions, such as formation of CO and/or H₂.^{3–8} Only a handful of well-defined molecular catalysts have been shown to lead to the selective formate formation.^{9–17} The product selectivity is generally controlled by either the thermodynamics or kinetics of the reaction between reduced metal and available substrates, namely protons and CO₂.^{14,18–21}

In natural photosynthesis, CO₂ is first captured from air and converted to 1,3-bisphosphoglycerate, which is then reduced via the hydride transfer (HT) from the NADPH cofactor.²² Following in nature's footsteps, our group recently developed hydrides that mimic NADPH activity and that are sufficiently hydridic to perform a direct reduction of CO₂.^{23–28} Our work has shown how the thermodynamic hydricities of metal-free hydrides can be tuned by using aromatization, delocalization, and hyperconjugation mechanisms and that the benzimidazole-based hydride C4H (Figure 1) can efficiently reduce CO₂ to formate via the HT mechanism.²⁶ We also demonstrated that the benzimidazole-based hydrides can be efficiently regenerated electrochemically by optimizing the conditions for proton-coupled electron transfer, reporting the first proof-of-

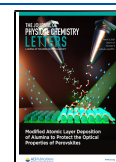
principle demonstration toward metal-free and selective CO₂ reduction catalysis.^{24,27} Despite the successful demonstration of HT to CO₂, the kinetics of the transformation were quite slow in our studies on the time scale of several hours.

Herein, we report the results of our investigation of the factors that control the kinetics of HT from metal-free hydrides to CO₂ by determining, both experimentally and computationally, the HT barriers of the series of hydride donors shown in Figure 1. As expected, our studies reveal that HT kinetics are improved by increasing the thermodynamic driving force of the reaction. The experimental data were fit to the Marcus hydride transfer model to obtain the relevant self-exchange reorganization energies of the metal-free hydrides and CO₂/HCO₂[−] couple. Our results show that the self-exchange reorganization energies of metal-free hydrides are relatively large (98 kcal/mol), indicating that the appropriate modification of the hydride donor structure yields lower reorganization energies and improved kinetics for HT. However, we find that even if optimal hydride donors were used, the HT to CO₂ would still be relatively slow due to the large self-exchange reorganization energy associated with the CO₂/HCO₂[−] couple (49.2 kcal/mol). Our results provide an explanation for why the formate

Received: December 12, 2020

Accepted: February 24, 2021

Published: March 2, 2021



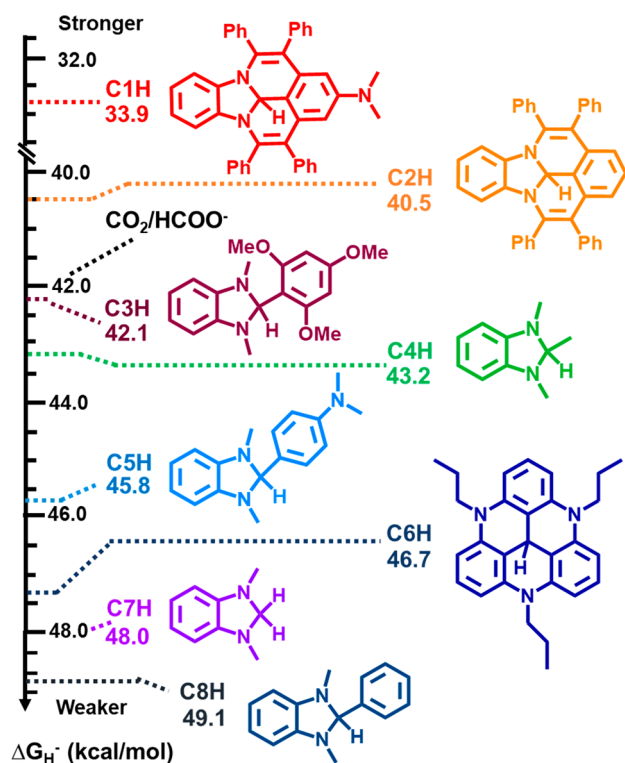


Figure 1. Structures and thermodynamic hydricities of hydride donors in DMSO. The hydricities were calculated at the wB97x/D 6-311+G(d,p) level of theory (computational details are provided in the Supporting Information, section S2). The hydricity of formate was obtained from the literature.²⁹

ion is a rarely observed product in electrocatalytic CO₂ reduction and point to possible approaches that can be used to circumvent the limitations of direct HT mechanism.

The synthesis and characterization of model compounds C1H to C8H are described in section S1 of the Supporting Information. The driving force for HT from organic hydrides to CO₂ was evaluated as the difference between the thermodynamic hydricities (ΔG_{H}^-) of organic hydrides and of formate ($\Delta G_{\text{H}}^- = 42 \text{ kcal/mol}^{29}$) in DMSO. The ΔG_{H}^- values for organic hydrides were calculated by using the DFT methodology that was previously shown to agree with

experimental values²⁴ (see Supporting Information, section S2, for computational details). Our results indicate that the strength of hydride donors can be improved in structures where the benzimidazole and aryl moieties are coplanar (C1H and C2H) or by introducing electron-donating amino and methoxy groups (C1H, C3H, and C5H). Overall, the driving force for HT from these hydrides to CO₂ varies between -11 and $+5 \text{ kcal/mol}$ (Table S2). The HT driving forces are close to $\Delta G = 0$, which is the optimal range for applications in electrocatalysis (if ΔG values were more negative, HT kinetics would likely improve, but the electrochemical regeneration of the hydride would require larger overpotentials).^{27,28}

Experimental reaction conditions were tested for CO₂ reduction by organo-hydrides to maximize the formate yields with respect to temperature, pressure, and solvent (Supporting Information, section S1C). Optimal formate production was achieved at high temperatures (100 °C), at mild CO₂ pressure ($\sim 2.2 \text{ atm}$), and in a polar medium (obtained by adding potassium salts to DMSO). An increase in the temperature led to an increase in formate yields (Figure S1) consistent with the large kinetic barrier associated with HT. The increased polarity of the medium, either by solvents with higher dielectric constants^{30,31} (Figures S2 and S3) or by adding alkali and alkaline salts (Figure S4), leads to an increase in formate yields as expected for a reaction that produced charged products from neutral by stabilizing the transition state and products.^{13,26,32,33} The solvent can also influence the kinetics of the hydride transfer process by modulating the rate of organization of reactants and solvent molecules into the configuration that precedes HT.^{34–37} Similarly, the cations from added salt may lower the transition state energy by stabilizing the negative charge that develops on oxygen atoms of CO₂.^{38–40} In all reactions, the loss of organic hydride reactants was higher than the growth of formate product (blue and red traces in Figures S1–S5), indicating additional reaction pathways that organic hydrides undergo. While the detailed investigation of this side reaction was not performed, we hypothesize that it involves the reduction of DMSO to DMS in the presence of a proton source based on the observed NMR peak at 2.0 ppm (Figures S6–S8). Additionally, the calculated reaction free energies between DMSO and organo-hydrides were found to be extremely favorable (see section

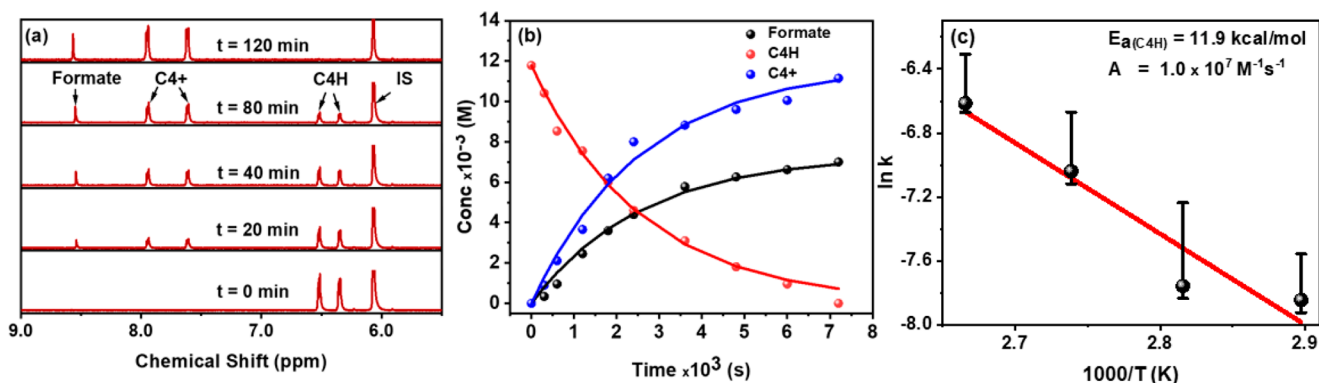


Figure 2. (a) ¹H NMR spectra of a reaction mixture containing C4H and CO₂ in DMSO-*d*₆. Reaction conditions: [C4H] = 0.01 M, [KBF₄] = 0.20 M, *P*_{CO₂} = $\sim 2.2 \text{ atm}$, *t* = 100 °C, and reaction times of 0, 20, 40, 80, and 120 min. 1,3,5-Trimethoxybenzene (0.05 M) was introduced as an internal NMR standard. (b) Scatter plot of reactant and product concentrations as a function of time for the reaction between C4H (0.1 M) and CO₂ (1.5 M). The solid lines represent the fits obtained by using the kinetic model described in section S1C of the Supporting Information. (c) Plot of $\ln k$ vs $1/T$ for the reaction between C4H and CO₂. Error bars represent the 95% confidence.

S2.B, Supporting Information), further confirming that the side reaction may involve DMSO reduction.

We next analyzed the kinetics of HT reactions using organohydrides at the optimized conditions by ^1H NMR spectroscopy. Figure 2 illustrates the kinetic analysis for C4H, while the results for other hydrides are shown in Table S2 and Figures S9–S16. Peaks associated with the hydride donor C4H (at 6.52 and 6.35 ppm) were found to decrease in intensity, while a concomitant increase in the formate (at 8.5 ppm) and the conjugate hydride acceptor C4^+ (at 7.94 and 7.61 ppm) peaks were observed (Figure 2a). The increase in the C4^+ concentration does not match the increase in the formate ion concentration (blue and black traces in Figure 2b), indicating that an additional HT pathway occurs (possibly reduction of DMSO, as discussed in the previous paragraph). Thus, the experimental HT data were fit to kinetic models that account for the side HT reactions.

Relevant rate constants were obtained by performing simultaneous fitting of concentration–time profiles for organic hydride, organic cation, and formate anion. The kinetic model used to fit the data includes the reaction of interest (shown in eq 1) and a side reaction involving HT to DMSO. The kinetic model provided a set of differential equations that were solved numerically, and experimental concentration profiles were fit by using rate constants as only the fitting parameters (model details are available in section S1C of the Supporting Information). The kinetic data fitting enabled the derivation of rate constants k_{HT} for HT from hydride donors (R–H) to CO_2 .



The resulting rate constant values are in the 2×10^{-6} – $2 \times 10^{-4} \text{ M}^{-1} \text{ s}^{-1}$ range (Table S2). Subsequent temperature-dependent kinetic experiments were performed to determine free energy barriers $\Delta G^\ddagger$ and the Arrhenius pre-exponential factors for HT reactions. These experiments were performed for C4H, C5H, and C7H (Figure 2c and Figure S17a,b), while other hydrides could not be measured because of the narrow temperature window in which detectable HT to CO_2 was observed. The free energy barriers for other hydride donors were thus evaluated by taking an average pre-exponential factor of $1 \times 10^7 \text{ M}^{-1} \text{ s}^{-1}$ (which was obtained from C4H, C5H, and C7H temperature-dependent kinetic experiments). Using this pre-exponential factor, we found the experimental free energy barriers to range from 15 to 21 kcal/mol (Table S2).

The experimentally determined free energy barriers are in good agreement with DFT-computed values (Table S2 and section S2, Supporting Information), except for C1H and C2H, where experiment yields much larger barriers than the computationally predicted values. We associate this mismatch to the poor solubility of these hydrides in DMSO, which results in aggregation and low reactivity observed in the experiment. The optimized transition state structures of R–H– CO_2 (Figure S21) demonstrate that CO_2 aligns parallel to the benzimidazole C–N bond, where the partial negative charge on O is stabilized by far-ranging interactions ($<3.5 \text{ \AA}$) with the N of the five-member ring. Faster kinetics can be correlated to several structural properties, such as shorter R–H bond lengths, longer H– CO_2 bond lengths, and more linear HT (C–H–C) angles (Table S8) and all factors indicating an early transition state.

The Marcus HT formalism^{41–43} was applied to the observed rate dependence on the thermodynamic driving force (ΔG^0),

with the goal of evaluating the self-exchange reorganization energies associated with HT to CO_2 . According to Marcus theory, the reaction barrier $\Delta G^\ddagger$ can be expressed as

$$\Delta G^\ddagger = W_r + \frac{\lambda}{4} + \frac{\Delta G^{0'}}{2} + \frac{\Delta G^{0'^2}}{4\lambda} \quad (2)$$

where W_r is the free energy for the formation of the precursor complex and $\Delta G^{0'}$ is the thermodynamic driving force (see section S1.C.III, Supporting Information). The reorganization energy λ is the energy required to distort the nuclei of the product state from the initial reactant geometry to the minimum product geometry. While previous studies have shown that the reorganization energy is structure-dependent,^{44,45} we decided to treat it here as a constant to keep the model simple and minimize the number of fitting parameters. Our approach is expected to introduce an error of 15% since the previously reported intrinsic barriers for structurally different organic hydrides tend to vary in the 85–100 kcal/mol range.⁴⁵ This error, while not insignificant, does not change general conclusions drawn in this Letter. It is also important to comment here that the Marcus theory for group transfer reactions does not account for tunneling effects. Even though tunneling is expected to be significant for HT processes, the use of Marcus formalism was shown previously to provide good agreement with the experimental kinetic isotope effects, mostly because tunneling effects fall off sharply as $\Delta G^{0'}$ moves away from the zero value.^{46–48} The experimental and DFT-calculated barriers are plotted in Figure 3a along with the fits obtained by using eq 2. The fit enabled us

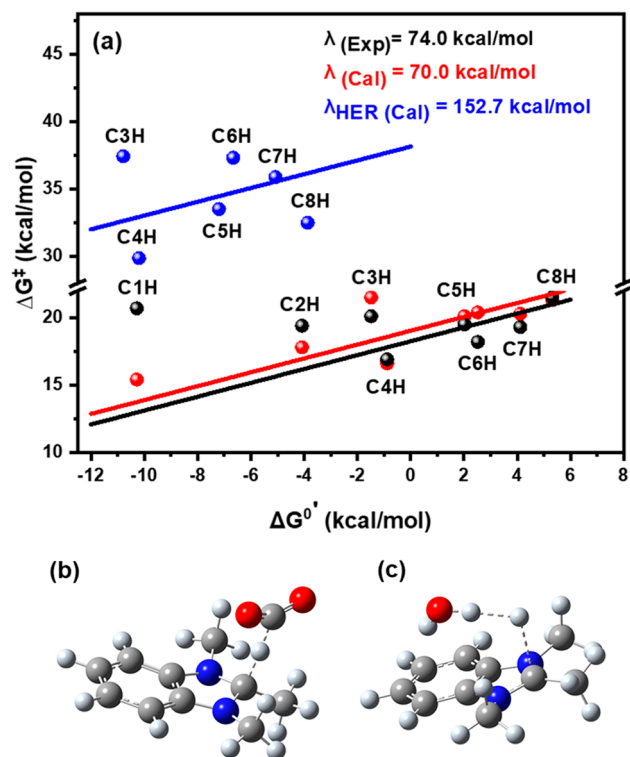


Figure 3. (a) Correlation between free energy barriers ($\Delta G^\ddagger$) and reaction free energies (ΔG^0) for HT between C1H–C8H and CO_2 : experiment (black), calculation (red). Blue data show calculated values for HT to water (HER). Solid lines represent the fit to the Marcus model. (b, c) Calculated TS structures for HT from C4H to (b) CO_2 and (c) H_2O .

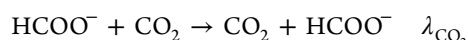
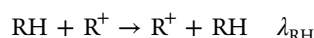
to evaluate the reorganization energy associated with HT from R–H to CO₂, and the values of 74 and 70 kcal/mol for experimental and calculated barriers were obtained.

Before we proceed to further discuss the analysis of reaction kinetics involving CO₂, we briefly discuss the reactivity of organic hydrides toward protons. While the hydrogen-evolution reaction (HER) is often the most challenging competing process to avoid in electrocatalytic CO₂ reduction, we found that our organic hydrides reduce protons to H₂ in negligible yields (<2%, section S1C.IV, [Supporting Information](#)). These experimental results are consistent with our DFT-calculated barriers for water reduction by HT from C1H–C8H, which were computed to be in the 26–32 kcal/mol range ([Figures 3a and 3c](#) and [Supporting Information](#) section S1C.IV).

Returning to CO₂ reduction kinetics, we further investigate different contributions to the reorganization energy λ . Specifically, λ can be expressed in terms of symmetric self-exchange reorganization energies as

$$\lambda = \frac{\lambda_{\text{RH}} + \lambda_{\text{CO}_2}}{2} \quad (3)$$

The reorganization energies λ_{RH} and λ_{CO_2} are associated with the following self-exchange reactions:



To estimate the $\lambda_{\text{R–H}}$ for our organo-hydrides, the self-exchange reorganization energy for C4H was obtained experimentally to be $\lambda_{\text{RH}} = 98$ kcal/mol ([Supporting Information](#), section S1C.III(a)). This value is in excellent agreement with previously reported self-exchange energies of organic hydrides by Kreevoy and co-workers.^{43,45,49,50} Finally, λ and λ_{RH} values were used to derive λ_{CO_2} , and the value of 49.2 kcal/mol was obtained, which is in reasonable agreement with the DFT-calculated value ($\lambda_{\text{CO}_2} = 56.4$ kcal/mol; see section S2.C, [Supporting Information](#)). A large λ_{CO_2} value obtained in our study is consistent with the previously reported self-exchange reorganization energy associated with the electron transfer involving the CO₂/CO₂^{•−} couple.⁵¹ In the case of electron transfer, the λ value was found to be 30.5 kcal/mol, and most of the inner-sphere contribution to this energy was assigned to a large change in the O–C–O bond angle upon one-electron reduction. Similar changes are expected to take place in our HT involving the CO₂/HCOO[−] couple and are likely the reason for the observed reorganization energy.

What information can be derived from these reorganization energies? To address this question, we evaluate the rate of HT for a hypothetical electrocatalyst operating at room temperature and at the thermodynamic driving force of −0.1 eV (to ensure a small overpotential). If the catalyst was made from our organic hydrides, the HT rate would be $2.5 \times 10^{-6} \text{ M}^{-1} \text{ s}^{-1}$ ([Table 1](#)), which is prohibitively slow. The only way to reduce the barrier would involve identifying hydride donors with lower self-exchange reorganization energies. Indeed, metal-based hydride donors tend to have lower self-exchange reorganization energies. For example, a Ru-based hydride [Ru(trpy)(bpy)H]⁺ was reported to have $\lambda_{\text{RH}} = 60.0$ kcal/mol,⁵² which lowers our rate constant estimate to $8.0 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$ ([Table 1](#)), a value that is still quite low. Interestingly,

Table 1. Summary of HT Rate Constants Estimated (Using Marcus and Arrhenius Equations) for Reduction of CO₂ at 25 °C with −0.1 eV Thermodynamic Driving Force^a

hydride donor	λ_{RH} (kcal/mol)	λ (kcal/mol)	k_{HT} (M ^{−1} s ^{−1})
NADH analogue	98.4	74.0	2.5×10^{-6}
[Ru(trpy)(bpy)H] ⁺	60.0	54.6	8.0×10^{-3}
ideal	0	24.6	3.0×10^3

^aThe estimation was made by using the experimentally obtained $\lambda_{\text{CO}_2} = 49.2$ kcal/mol value.

even if an ideal hydride donor was identified, with $\lambda_{\text{RH}} = 0$ kcal/mol, the rate constant would be relatively small ($3.0 \times 10^3 \text{ M}^{-1} \text{ s}^{-1}$, [Table 1](#)). This rate constant is barely sufficient for processes involving the use of solar energy, where catalytic rates in excess of 100 turnovers per second are needed to efficiently keep up with the photon flux at solar irradiance levels.^{53,54} Also, rate constants listed in [Table 1](#) provide a plausible explanation for the fact that electrocatalytic CO₂ reduction to the formate ion is rarely observed experimentally. This upper limits for HT rate constants at mild driving forces are fully consistent with previous reports involving metal-based hydride donors.^{55–59} For example, several careful studies of Ru-based hydride donors have enabled a direct comparison of HT rate constants and thermodynamic driving forces.^{57–59} The highest rate constant among these studies was reported to be $29.7 \text{ M}^{-1} \text{ s}^{-1}$ (for the driving force of −10.5 kcal/mol), which approaches, but does not exceed, the $10^3 \text{ M}^{-1} \text{ s}^{-1}$ upper limit that we hypothesize in this Letter. Furthermore, a recent report of electrocatalytic CO₂ reduction to formate by a Pt-based hydride catalyst has shown excellent product selectivity at mild overpotentials but exhibited sluggish reaction kinetics,⁵⁵ which is fully consistent with the conclusions drawn in our paper.

What can be done to circumvent the observed kinetic barriers for direct HT to CO₂? An interesting approach has been recently proposed by Yang, namely, the orthogonal HT mechanism that involves a separate transfer of electrons and a proton to CO₂.⁶⁰ In this bioinspired approach, the hydride-donating catalyst is replaced by a two-component catalytic system consisting of separate electron and proton donating groups. Using thermodynamic arguments, the authors show that the orthogonal HT can be efficiently utilized to break the scaling relationships associated with conventional HT. The orthogonal HT could also provide an excellent solution to the kinetic challenges associated with conventional HT because the kinetic barriers associated with electron and proton transfers are often much lower than those for HTs.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.0c03662>.

General methods, synthesis, hydride transfer studies, and computational methods ([PDF](#))

■ AUTHOR INFORMATION

Corresponding Author

Ksenija D. Glusac – Department of Chemistry, University of Illinois at Chicago, Chicago, Illinois 60607, United States; Chemical Sciences and Engineering Division, Argonne National Laboratory, Lemont, Illinois 60439, United States;

orcid.org/0000-0002-2734-057X; Email: glusac@uic.edu

Authors

Ravindra B. Weerasooriya – Department of Chemistry, University of Illinois at Chicago, Chicago, Illinois 60607, United States; Chemical Sciences and Engineering Division, Argonne National Laboratory, Lemont, Illinois 60439, United States

Jonathan L. Gesiorski – Department of Chemistry, University of Illinois at Chicago, Chicago, Illinois 60607, United States; Chemical Sciences and Engineering Division, Argonne National Laboratory, Lemont, Illinois 60439, United States

Abdulaziz Alherz – Department of Chemical and Biological Engineering, University of Colorado Boulder, Boulder, Colorado 80309, United States; orcid.org/0000-0001-7529-3483

Stefan Ilic – Department of Chemistry, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061, United States; orcid.org/0000-0002-6305-4001

George N. Hargenrader – Department of Chemistry, University of Illinois at Chicago, Chicago, Illinois 60607, United States; Chemical Sciences and Engineering Division, Argonne National Laboratory, Lemont, Illinois 60439, United States

Charles B. Musgrave – Department of Chemical and Biological Engineering, University of Colorado Boulder, Boulder, Colorado 80309, United States; Department of Chemistry, University of Colorado Boulder, Boulder, Colorado 80309, United States; orcid.org/0000-0002-5732-3180

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.jpclett.0c03662>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Science Foundation (1954298 and 1565971).

REFERENCES

- (1) Mellmann, D.; Sponholz, P.; Junge, H.; Beller, M. Formic Acid as a Hydrogen Storage Material—Development of Homogeneous Catalysts for Selective Hydrogen Release. *Chem. Soc. Rev.* **2016**, *45* (14), 3954–3988.
- (2) Bi, J.; Hou, P.; Kang, P. Single Iridium Pincer Complex for Roundtrip Electrochemical Conversion between Carbon Dioxide and Formate. *ChemCatChem* **2019**, *11* (8), 2069–2072.
- (3) Zhanaidarova, A.; Jones, S. C.; Despagnet-Ayoub, E.; Pimentel, B. R.; Kubiak, C. P. Re (TBU-Bpy)(CO)₃Cl Supported on Multi-Walled Carbon Nanotubes Selectively Reduces CO₂ in Water. *J. Am. Chem. Soc.* **2019**, *141* (43), 17270–17277.
- (4) Chen, Z.; Chen, C.; Weinberg, D. R.; Kang, P.; Concepcion, J. J.; Harrison, D. P.; Brookhart, M. S.; Meyer, T. J. Electrocatalytic Reduction of CO₂ to CO by Polypyridyl Ruthenium Complexes. *Chem. Commun.* **2011**, *47* (47), 12607–12609.
- (5) Chen, Z.; Kang, P.; Zhang, M.-T.; Meyer, T. J. Making Syngas Electrocatalytically Using a Polypyridyl Ruthenium Catalyst. *Chem. Commun.* **2014**, *50* (3), 335–337.
- (6) Windle, C. D.; Pastor, E.; Reynal, A.; Whitwood, A. C.; Vaynzof, Y.; Durrant, J. R.; Perutz, R. N.; Reisner, E. Improving the Photocatalytic Reduction of CO₂ to CO through Immobilisation of a Molecular Re Catalyst on TiO₂. *Chem. - Eur. J.* **2015**, *21* (9), 3746–3754.
- (7) Woolerton, T. W.; Sheard, S.; Reisner, E.; Pierce, E.; Ragsdale, S. W.; Armstrong, F. A. Efficient and Clean Photoreduction of CO₂ to CO by Enzyme-Modified TiO₂ Nanoparticles Using Visible Light. *J. Am. Chem. Soc.* **2010**, *132* (7), 2132–2133.
- (8) Kuehnle, M. F.; Orchard, K. L.; Dalle, K. E.; Reisner, E. Selective Photocatalytic CO₂ Reduction in Water through Anchoring of a Molecular Ni Catalyst on CdS Nanocrystals. *J. Am. Chem. Soc.* **2017**, *139* (21), 7217–7223.
- (9) Kang, P.; Cheng, C.; Chen, Z.; Schauer, C. K.; Meyer, T. J.; Brookhart, M. Selective Electrocatalytic Reduction of CO₂ to Formate by Water-Stable Iridium Dihydride Pincer Complexes. *J. Am. Chem. Soc.* **2012**, *134* (12), 5500–5503.
- (10) Waldie, K. M.; Brunner, F. M.; Kubiak, C. P. Transition Metal Hydride Catalysts for Sustainable Interconversion of CO₂ and Formate: Thermodynamic and Mechanistic Considerations. *ACS Sustainable Chem. Eng.* **2018**, *6* (5), 6841–6848.
- (11) Ceballos, B. M.; Yang, J. Y. Highly Selective Electrocatalytic CO₂ Reduction by [Pt (Dmpe)₂]²⁺ through Kinetic and Thermodynamic Control. *Organometallics* **2020**, *39* (9), 1491–1496.
- (12) Margarit, C. G.; Asimow, N. G.; Costentin, C.; Nocera, D. G. Tertiary Amine-Assisted Electoreduction of Carbon Dioxide to Formate Catalyzed by Iron Tetraphenylporphyrin. *ACS Energy Lett.* **2020**, *5* (1), 72–78.
- (13) Roy, S.; Sharma, B.; Pécaut, J.; Simon, P.; Fontecave, M.; Tran, P. D.; Derat, E.; Artero, V. Molecular Cobalt Complexes with Pendant Amines for Selective Electrocatalytic Reduction of Carbon Dioxide to Formic Acid. *J. Am. Chem. Soc.* **2017**, *139* (10), 3685–3696.
- (14) Ceballos, B. M.; Yang, J. Y. Directing the Reactivity of Metal Hydrides for Selective CO₂ Reduction. *Proc. Natl. Acad. Sci. U. S. A.* **2018**, *115* (50), 12686–12691.
- (15) Loewen, N. D.; Neelakantan, T. V.; Berben, L. A. Renewable Formate from C–H Bond Formation with CO₂: Using Iron Carbonyl Clusters as Electrocatalysts. *Acc. Chem. Res.* **2017**, *50* (9), 2362–2370.
- (16) Taheri, A.; Berben, L. A. Tailoring Electrocatalysts for Selective CO₂ or H⁺ Reduction: Iron Carbonyl Clusters as a Case Study. *Inorg. Chem.* **2016**, *55* (2), 378–385.
- (17) Taheri, A.; Thompson, E. J.; Fetting, J. C.; Berben, L. A. An Iron Electrocatalyst for Selective Reduction of CO₂ to Formate in Water: Including Thermochemical Insights. *ACS Catal.* **2015**, *5* (12), 7140–7151.
- (18) Ramakrishnan, S.; Moretti, R. A.; Chidsey, C. E. D. Mapping Free Energy Regimes in Electrocatalytic Reductions to Screen Transition Metal-Based Catalysts. *Chem. Sci.* **2019**, *10* (32), 7649–7658.
- (19) Waldie, K. M.; Ostericher, A. L.; Reineke, M. H.; Sasayama, A. F.; Kubiak, C. P. Hydricity of Transition-Metal Hydrides: Thermodynamic Considerations for CO₂ Reduction. *ACS Catal.* **2018**, *8* (2), 1313–1324.
- (20) Barlow, J. M.; Yang, J. Y. Thermodynamic Considerations for Optimizing Selective CO₂ Reduction by Molecular Catalysts. *ACS Cent. Sci.* **2019**, *5* (4), 580–588.
- (21) Johnson, S. I.; Nielsen, R. J.; Goddard, W. A., III Selectivity for HCO₂[−] over H₂ in the Electrochemical Catalytic Reduction of CO₂ by (POCOP) IrH₂. *ACS Catal.* **2016**, *6* (10), 6362–6371.
- (22) Schneider, G.; Lindqvist, Y.; Branden, C.-I. RUBISCO: Structure and Mechanism. *Annu. Rev. Biophys. Biomol. Struct.* **1992**, *21* (1), 119–143.
- (23) Ilic, S.; Alherz, A.; Musgrave, C. B.; Glusac, K. D. Thermodynamic and Kinetic Hydricities of Metal-Free Hydrides. *Chem. Soc. Rev.* **2018**, *47* (8), 2809–2836.
- (24) Ilic, S.; Pandey Kadel, U.; Basdogan, Y.; Keith, J. A.; Glusac, K. D. Thermodynamic Hydricities of Biomimetic Organic Hydride Donors. *J. Am. Chem. Soc.* **2018**, *140* (13), 4569–4579.
- (25) Hargenrader, G. N.; Weerasooriya, R. B.; Ilic, S.; Niklas, J.; Poluektov, O. G.; Glusac, K. D. Photoregeneration of Biomimetic Nicotinamide Adenine Dinucleotide Analogues via a Dye-Sensitized Approach. *ACS Appl. Energy Mater.* **2019**, *2* (1), 80–91.
- (26) Lim, C.-H.; Ilic, S.; Alherz, A.; Worrell, B. T.; Bacon, S. S.; Hynes, J. T.; Glusac, K. D.; Musgrave, C. B. Benzimidazoles as Metal-

Free and Recyclable Hydrides for CO₂ Reduction to Formate. *J. Am. Chem. Soc.* **2019**, *141* (1), 272–280.

(27) Ilic, S.; Alherz, A.; Musgrave, C. B.; Glusac, K. D. Importance of Proton-Coupled Electron Transfer in Cathodic Regeneration of Organic Hydrides. *Chem. Commun.* **2019**, 55 (39), 5583–5586.

(28) Lim, C.-H.; Holder, A. M.; Hynes, J. T.; Musgrave, C. B. Reduction of CO₂ to Methanol Catalyzed by a Biomimetic Organo-Hydride Produced from Pyridine. *J. Am. Chem. Soc.* **2014**, *136* (45), 16081–16095.

(29) Tsay, C.; Yang, J. Y. Electrocatalytic Hydrogen Evolution under Acidic Aqueous Conditions and Mechanistic Studies of a Highly Stable Molecular Catalyst. *J. Am. Chem. Soc.* **2016**, *138* (43), 14174–14177.

(30) Wiedner, E. S.; Linehan, J. C. Making a Splash in Homogeneous CO₂ Hydrogenation: Elucidating the Impact of Solvent on Catalytic Mechanisms. *Chem. - Eur. J.* **2018**, *24* (64), 16964–16971.

(31) Creutz, C.; Chou, M. H. Rapid Transfer of Hydride Ion from a Ruthenium Complex to C1 Species in Water. *J. Am. Chem. Soc.* **2007**, *129* (33), 10108–10109.

(32) Boston, D. J.; Xu, C.; Armstrong, D. W.; MacDonnell, F. M. Photochemical Reduction of Carbon Dioxide to Methanol and Formate in a Homogeneous System with Pyridinium Catalysts. *J. Am. Chem. Soc.* **2013**, *135* (44), 16252–16255.

(33) Qiao, J.; Liu, Y.; Hong, F.; Zhang, J. A Review of Catalysts for the Electroreduction of Carbon Dioxide to Produce Low-Carbon Fuels. *Chem. Soc. Rev.* **2014**, *43* (2), 631–675.

(34) Kreevoy, M. M.; Kotchevar, A. T. Dynamics of Hydride Transfer between NAD⁺ Analogs. *J. Am. Chem. Soc.* **1990**, *112* (9), 3579–3583.

(35) Kotchevar, A. T.; Kreevoy, M. M. Solvent Modulation of the Dynamics of Hydride Transfer. *J. Phys. Chem.* **1991**, *95* (25), 10345–10350.

(36) Sullivan, B. P.; Meyer, T. J. Kinetics and Mechanism of Carbon Dioxide Insertion into a Metal-Hydride Bond. A Large Solvent Effect and an Inverse Kinetic Isotope Effect. *Organometallics* **1986**, *5* (7), 1500–1502.

(37) Taheri, A.; Carr, C. R.; Berben, L. A. Electrochemical Methods for Assessing Kinetic Factors in the Reduction of CO₂ to Formate: Implications for Improving Electrocatalyst Design. *ACS Catal.* **2018**, *8* (7), 5787–5793.

(38) Bhugun, I.; Lexa, D.; Savéant, J.-M. Catalysis of the Electrochemical Reduction of Carbon Dioxide by Iron (0) Porphyrins. Synergistic Effect of Lewis Acid Cations. *J. Phys. Chem.* **1996**, *100* (51), 19981–19985.

(39) Hong, D.; Kawanishi, T.; Tsukakoshi, Y.; Kotani, H.; Ishizuka, T.; Kojima, T. Efficient Photocatalytic CO₂ Reduction by a Ni (II) Complex Having Pyridine Pendants through Capturing a Mg²⁺ Ion as a Lewis-Acid Cocatalyst. *J. Am. Chem. Soc.* **2019**, *141* (51), 20309–20317.

(40) Heimann, J. E.; Bernskoetter, W. H.; Hazari, N. Understanding the Individual and Combined Effects of Solvent and Lewis Acid on CO₂ Insertion into a Metal Hydride. *J. Am. Chem. Soc.* **2019**, *141* (26), 10520–10529.

(41) Mayer, J. M. Simple Marcus-Theory-Type Model for Hydrogen-Atom Transfer/Proton-Coupled Electron Transfer. *J. Phys. Chem. Lett.* **2011**, *2* (12), 1481–1489.

(42) Brinkley, D. W.; Roth, J. P. Determination of a Large Reorganization Energy Barrier for Hydride Abstraction by Glucose Oxidase. *J. Am. Chem. Soc.* **2005**, *127* (45), 15720–15721.

(43) Lee, I.-S. H.; Jeoung, E. H.; Kreevoy, M. M. Marcus Theory of a Parallel Effect on α for Hydride Transfer Reaction between NAD⁺ Analogues. *J. Am. Chem. Soc.* **1997**, *119* (11), 2722–2728.

(44) Richter, D.; Tan, Y.; Antipova, A.; Zhu, X.; Mayr, H. Kinetics of Hydride Abstractions from 2-Arylbenzimidazolines. *Chem.—Asian J.* **2009**, *4* (12), 1824–1829.

(45) Kreevoy, M. M.; Ostovic, D.; Lee, I. S. H.; Binder, D. A.; King, G. W. Structure Sensitivity of the Marcus. Lambda. for Hydride

Transfer between NAD⁺ Analogs. *J. Am. Chem. Soc.* **1988**, *110* (2), 524–530.

(46) Han Lee, I.-S.; Jeoung, E. H.; Kreevoy, M. M. Primary Kinetic Isotope Effects on Hydride Transfer from 1, 3-Dimethyl-2-Phenylbenzimidazoline to NAD⁺ Analogues. *J. Am. Chem. Soc.* **2001**, *123* (31), 7492–7496.

(47) Kreevoy, M. M.; Ostovic, D.; Truhlar, D. G.; Garrett, B. C. Phenomenological Manifestations of Large-Curvature Tunneling in Hydride-Transfer Reactions. *J. Phys. Chem.* **1986**, *90* (16), 3766–3774.

(48) Kim, Y.; Truhlar, D. G.; Kreevoy, M. M. An Experimentally Based Family of Potential Energy Surfaces for Hydride Transfer between NAD⁺ Analogs. *J. Am. Chem. Soc.* **1991**, *113* (21), 7837–7847.

(49) Roberts, R. M. G.; Ostović, D.; Kreevoy, M. M. Hydride Transfer between NAD⁺ Analogues. *Faraday Discuss. Chem. Soc.* **1982**, *74*, 257–265.

(50) Lee, I. S. H.; Ostovic, D.; Kreevoy, M. M. Marcus Theory of Hydride Transfer from an Anionic Reduced Deazaflavin to NAD⁺ Analogs. *J. Am. Chem. Soc.* **1988**, *110* (12), 3989–3993.

(51) Bennett, L. E.; Warlop, P. Electron Transfer to Ozone: Outer-Sphere Reactivities of the Ozone/Ozonide and Related Non-Metal Redox Couples. *Inorg. Chem.* **1990**, *29* (10), 1975–1981.

(52) Matsubara, Y.; Fujita, E.; Doherty, M. D.; Muckerman, J. T.; Creutz, C. Thermodynamic and Kinetic Hydricity of Ruthenium (II) Hydride Complexes. *J. Am. Chem. Soc.* **2012**, *134* (38), 15743–15757.

(53) Lewis, N. S.; Crabtree, G.; Nozik, A. J.; Wasielewski, M. R.; Alivisatos, P.; Kung, H.; Tsao, J.; Chandler, E.; Walukiewicz, W.; Spitler, M. *Basic Research Needs for Solar Energy Utilization. Report of the Basic Energy Sciences Workshop on Solar Energy Utilization*, April 18–21, 2005; DOE/SC (USDOE Office of Science (SC)), 2005.

(54) Arakawa, H.; Aresta, M.; Armor, J. N.; Barteau, M. A.; Beckman, E. J.; Bell, A. T.; Bercaw, J. E.; Creutz, C.; Dinjus, E.; Dixon, D. A.; et al. Catalysis Research of Relevance to Carbon Management: Progress, Challenges, and Opportunities. *Chem. Rev.* **2001**, *101* (4), 953–996.

(55) Cunningham, D. W.; Barlow, J. M.; Velazquez, R. S.; Yang, J. Y. Reversible and Selective CO₂ to HCO₂[−] Electrocatalysis near the Thermodynamic Potential. *Angew. Chem., Int. Ed.* **2020**, *59* (11), 4443–4447.

(56) Mondal, B.; Neese, F.; Ye, S. Control in the Rate-Determining Step Provides a Promising Strategy to Develop New Catalysts for CO₂ Hydrogenation: A Local Pair Natural Orbital Coupled Cluster Theory Study. *Inorg. Chem.* **2015**, *54* (15), 7192–7198.

(57) Ramakrishnan, S.; Waldie, K. M.; Warnke, I.; De Crisci, A. G.; Batista, V. S.; Waymouth, R. M.; Chidsey, C. E. D. Experimental and Theoretical Study of CO₂ Insertion into Ruthenium Hydride Complexes. *Inorg. Chem.* **2016**, *55* (4), 1623–1632.

(58) Konno, H.; Kobayashi, A.; Sakamoto, K.; Fagalde, F.; Katz, N. E.; Saitoh, H.; Ishitani, O. Synthesis and Properties of [Ru(Tpy)(4,4'-X2bpy)H]⁺ (Tpy= 2,2':6',2''-Terpyridine, Bpy= 2, 2'-Bipyridine, X= H and MeO), and Their Reactions with CO₂. *Inorg. Chim. Acta* **2000**, *299* (2), 155–163.

(59) Sampaio, R. N.; Grills, D. C.; Polyansky, D. E.; Szalda, D. J.; Fujita, E. Unexpected Roles of Triethanolamine in the Photochemical Reduction of CO₂ to Formate by Ruthenium Complexes. *J. Am. Chem. Soc.* **2020**, *142* (5), 2413–2428.

(60) Yang, J. Y.; Kerr, T. A.; Wang, X. S.; Barlow, J. M. Reducing CO₂ to HCO₂[−] at Mild Potentials: Lessons from Formate Dehydrogenase. *J. Am. Chem. Soc.* **2020**, *142*, 19438.